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Brijesh Kumar Chaurasia, Nikita Tiwari & Nikhil Pateria

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Privacy-Preserving Vaccine Supply Chain Management Leveraging Blockchain and Self-Sovereign Identity

Brijesh Kumar Chaurasia¹, Nikita Tiwari^{2*}, and Nikhil Pateria^{3*}

¹ Department of Computer Science and Engineering,
Pranveer Singh Institute of Technology, Kanpur, UP, INDIA

²School of Computer Science,
University of Petroleum and Energy Studies, Dehradun, UK INDIA

³Department of AI & ML,
Manipal University Jaipur, Jaipur, Rajasthan, INDIA

*Corresponding Author

*nikita.tiwari25@gmail.com

*nikhil.pateria@jaipur.manipal.edu

Abstract. Blockchain technology's potential to enhance efficiency, security, and transparency has attracted a lot of attention to its integration into vaccination supply chain management in recent years. This work presents a comprehensive study on leveraging self-sovereign identity (SSI), blockchain, and privacy-preserving techniques to optimize vaccine distribution. We conduct a comparative analysis of existing blockchain-based frameworks, evaluating their effectiveness in addressing key challenges such as traceability, counterfeit prevention, and data privacy. The empirical results prove the efficacy of the proposed supply chain framework. Additionally, we identify and discuss critical challenges current systems face, including scalability issues, interoperability constraints, and privacy concerns. Our findings provide a robust foundation for future research and development in the field, proposing potential improvements and strategies to overcome the identified challenges.

Keywords: Blockchain, Supply Chain Management, Self-Sovereign Identity (SSI), Traceability, Counterfeit Prevention, and Interoperability.

1 Introduction

The COVID-19 pandemic caused by the SARS-CoV-2 virus posed a major global public health challenge, placing unprecedented strain

on healthcare systems and medical supply chains worldwide [1]. In response, rapid vaccine development and large-scale immunization campaigns were initiated as critical mitigation measures [2,3]. However, alongside accelerated vaccine production and distribution, **drug and vaccine supply chain counterfeiting emerged as a serious threat**, leading to adverse health outcomes and undermining public trust in vaccination programs [4]. Fig. 1 illustrates the exponential increase in reported counterfeit pharmaceutical incidents globally, with projections indicating continued growth beyond 2025 [5].

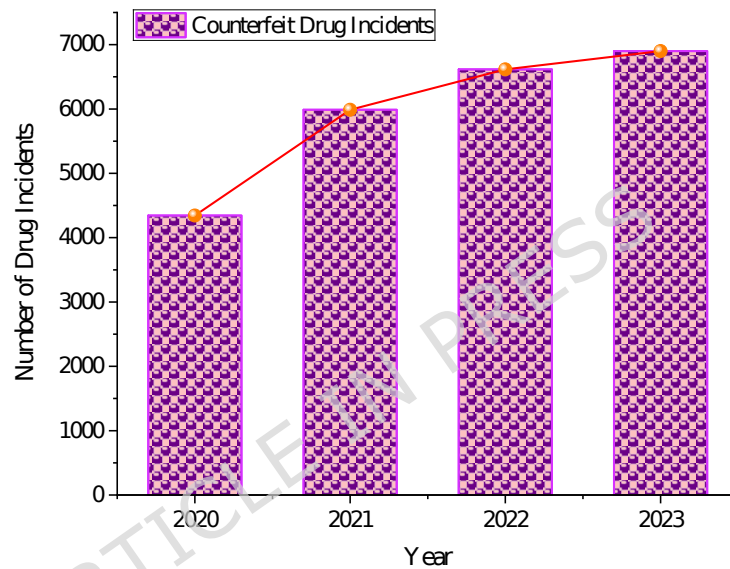


Fig. 1. Counterfeit Incidents Concerning Pharmaceuticals

In this context, **emerging technologies such as blockchain and Self-Sovereign Identity (SSI)** have gained attention for improving the efficiency, transparency, and security of vaccine supply chain management. Blockchain enables immutable tracking and verification of vaccine batches across distributed stakeholders, thereby strengthening trust and traceability [6]. The major benefits achievable through an efficient vaccine distribution framework are illustrated in **Fig. 2**.

- *Security & Fraud Prevention:* Public confidence is essential for successful vaccination programs. Transparent and verifiable vaccine distribution systems help prevent counterfeiting and strengthen trust, while delays and logistical gaps undermine immunization efforts [7]. Integrating blockchain, SSI, and QR-

based verification enhances traceability, accountability, and patient safety across the vaccine supply chain.

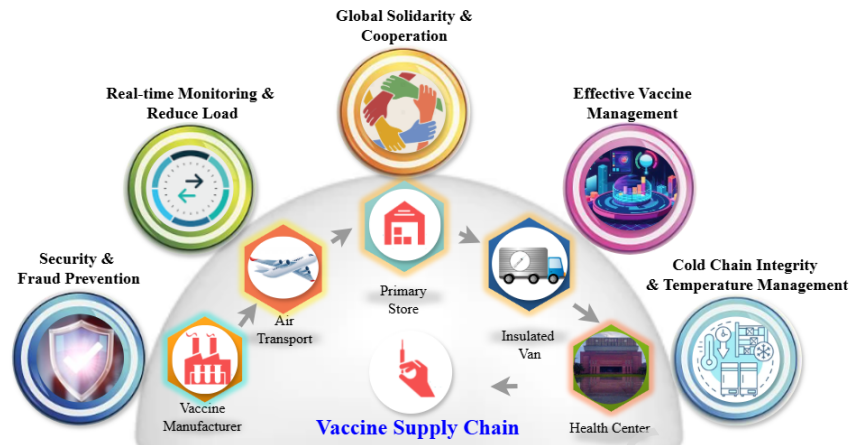


Fig. 2. Factors achieved with effective vaccine distribution system

- *Real-time Monitoring & Reduce Supply-chain Load*: Efficient vaccine distribution reduces healthcare system burden by minimizing delays, preventing shortages, and ensuring timely delivery to required locations [8]. Privacy-preserving vaccine supply chain management leveraging blockchain and SSI enables real-time monitoring, accurate inventory tracking, and improved resource utilization across the distribution network.
- *Global Solidarity & Cooperation*: Equitable vaccine distribution requires trusted cross-border collaboration, which is often constrained by data privacy, governance, and interoperability challenges. Integrating privacy-preserving blockchain and SSI mechanisms enables secure data sharing and trusted coordination across international vaccine supply chains [9].
- *Effective Vaccine Management*: Data-driven analytics enhance vaccine supply chain transparency and efficiency by enabling real-time monitoring, demand forecasting, and proactive risk mitigation [10]. In the context of drug traceability, predictive insights support timely interventions, reduce counterfeit exposure, and improve strategic planning, with AI and machine learning further strengthening decision-making accuracy [11].
- *Cold Chain Integrity & Temperature Management*: **Cold Chain Integrity & Temperature Management**: Maintaining controlled temperature conditions is essential for preserving vaccine efficacy and patient safety. IoT-enabled monitoring

systems support real-time tracking of storage and transit conditions, while integration with traceability platforms ensures regulatory compliance, operational visibility, and reduced product loss [12,13].

In view of these factors, the need for effective vaccine distribution cannot be overemphasized. Using innovative technology like blockchain provides an opportunity to expedite distribution processes, improve accountability, and strengthen healthcare systems in the face of future pandemics and public health emergencies. Blockchain technology provides several advantages that can considerably improve COVID-19 vaccine delivery methods. Blockchain has the potential to facilitate equitable and effective vaccine delivery by enhancing transparency, traceability, efficiency, and safety, ultimately helping global efforts to combat the epidemic and protect public health.

Novelty and Contributions Beyond Blockchain-Only Vaccine Supply Chains:

Existing blockchain-based vaccine supply chain solutions primarily focus on immutable record-keeping, transaction transparency, and batch-level traceability. While these approaches improve auditability and counterfeit detection, they largely rely on identity-agnostic or centrally managed identity models, where stakeholders are authenticated through predefined roles or platform-specific accounts. Such designs limit privacy preservation, cross-organizational interoperability, and fine-grained access control, particularly in multi-country and multi-regulator vaccine ecosystems.

The proposed framework advances the state of the art by **tightly integrating Self-Sovereign Identity (SSI) with blockchain-based traceability**, enabling decentralized, privacy-preserving, and verifiable identity management across the entire vaccine supply chain. Unlike blockchain-only systems, where identities are implicitly trusted or centrally provisioned, our approach allows each stakeholder—manufacturers, distributors, healthcare centers, and clinics to operate using **verifiable credentials (VCs) bound to decentralized identifiers (DIDs)** issued by trusted authorities.

A key novelty of the proposed system lies in **decoupling identity verification from on-chain transactions**. Identity validation, selective disclosure, and revocation checks are performed off-chain using SSI mechanisms, while the blockchain is used exclusively for anchoring cryptographic proofs, custody events, and immutable audit logs. This design significantly reduces on-chain data exposure,

ensures compliance with data protection regulations, and prevents identity linkability across transactions limitations commonly observed in identity-agnostic blockchain solutions.

Furthermore, the integration of SSI enables **selective disclosure and role-based authorization**, allowing stakeholders to prove eligibility (e.g., certified manufacturer, authorized distributor, licensed healthcare provider) without revealing unnecessary identity attributes. This capability is absent in conventional blockchain-based vaccine traceability frameworks, which typically expose full participant identities or rely on static access control lists.

By combining SSI with blockchain, the proposed framework supports **cross-border interoperability, revocable trust, and patient-centric privacy guarantees**, making it particularly suitable for large-scale vaccination programs operating across heterogeneous regulatory environments. To the best of our knowledge, this work represents one of the few comprehensive efforts that systematically integrate SSI with blockchain to address both **traceability and identity privacy** in vaccine supply chain management.

1.1 Motivation

In recent years, the pharmaceutical industry has been struggling with the pervasive issue of drug counterfeiting, which poses significant risks to public health and safety. The emergence of counterfeit drugs in the supply chain undermines consumer trust, leads to severe health consequences, and results in substantial financial losses for manufacturers and governments. With the global trade in counterfeit medicines being a multi-billion-dollar industry, there is an urgent need for robust traceability mechanisms that ensure the authenticity and integrity of pharmaceutical products from production to the end consumer [14]. This study is motivated by the critical need to explore, evaluate, and enhance drug traceability systems to effectively combat the growing threat of counterfeit drugs. Table 1 comprises a list of acronyms used in this work to make it simpler to read.

Table 1 List of common acronyms and descriptions used in the paper

S. No.	Abbreviation	Meaning
1	COVID	COronaVirus Disease
2	WHO	World Health Organization

3	SSI	Self-Sovereign Identity
4	SOTA	State-of-the-art
5	IPFS	InterPlanetary File System
6	SSI	Self-sovereign identity
7	DL	Distributed ledger
8	BT	Blockchain Technology
9	IoT	Internet of Things
10	RFID	Radio Frequency Identification
11	DApps	Decentralized Applications
12	AI	Artificial Intelligence
13	PoW	Proof of Work
14	PoS	Proof of Stake
15	PBFT	Practical Byzantine Fault Tolerance
16	VCs	Verifiable Credentials
17	DIDs	Decentralized Identifiers
18	W3C	World Wide Web Consortium
19	SCM	Supply Chain Management
20	EHRs	Electronic health records
21	GPS	Global positioning system
22	GDPR	General Data Protection Regulation
23	HIPAA	Health Insurance Portability and Accountability Act

1.2 Contribution

The contributions of the work are summarized as follows:

- The paper extensively studies existing drug traceability systems, offering a detailed analysis of their methodologies,

technologies, and efficacy in ensuring drug safety and authenticity.

- A thorough comparison of different supply chain management systems is presented, highlighting their strengths and weaknesses and identifying areas for improvement.
- The paper explores cutting-edge technologies, such as blockchain and SSI, that are being integrated into privacy-preserving traceability systems, offering novel insights into their potential to revolutionize drug traceability.
- The results show the efficacy of the proposed system. Additionally, the paper identifies emerging trends and future research directions in drug traceability, providing a roadmap for researchers and practitioners to address the challenges and opportunities in this critical area.
- We propose a **blockchain-SSI integrated vaccine supply chain framework** that overcomes the privacy, identity management, and interoperability limitations of blockchain-only and identity-agnostic traceability systems.

1.3 Structure of paper

This section describes the flow of the paper; it begins with an introduction to the problem of drug counterfeiting and the importance of drug traceability systems. It sets the context for highlighting the study's objectives, motivation, and contributions to the work. The detailed organization is illustrated in Fig. 3. Section 2 reviews the SSI & blockchain implications, focusing on privacy-preserving vaccine supply chain management and their applications in healthcare. Section 3 discusses the preliminaries of blockchain and SSI and their applications in various industries. Section 4 provides a detailed methodology of the proposed privacy-preserving vaccine supply chain management leveraging blockchain and SSI. Section 5 demonstrates the empirical results of the proposed framework with comparative analysis. Section 6 and Section 7 explore the challenges of blockchain-based frameworks for vaccine distribution and case studies, respectively. The paper discusses future research directions in Section 8, highlighting emerging technologies and methodologies that could further enhance drug privacy-preserved traceability, followed by a conclusion.

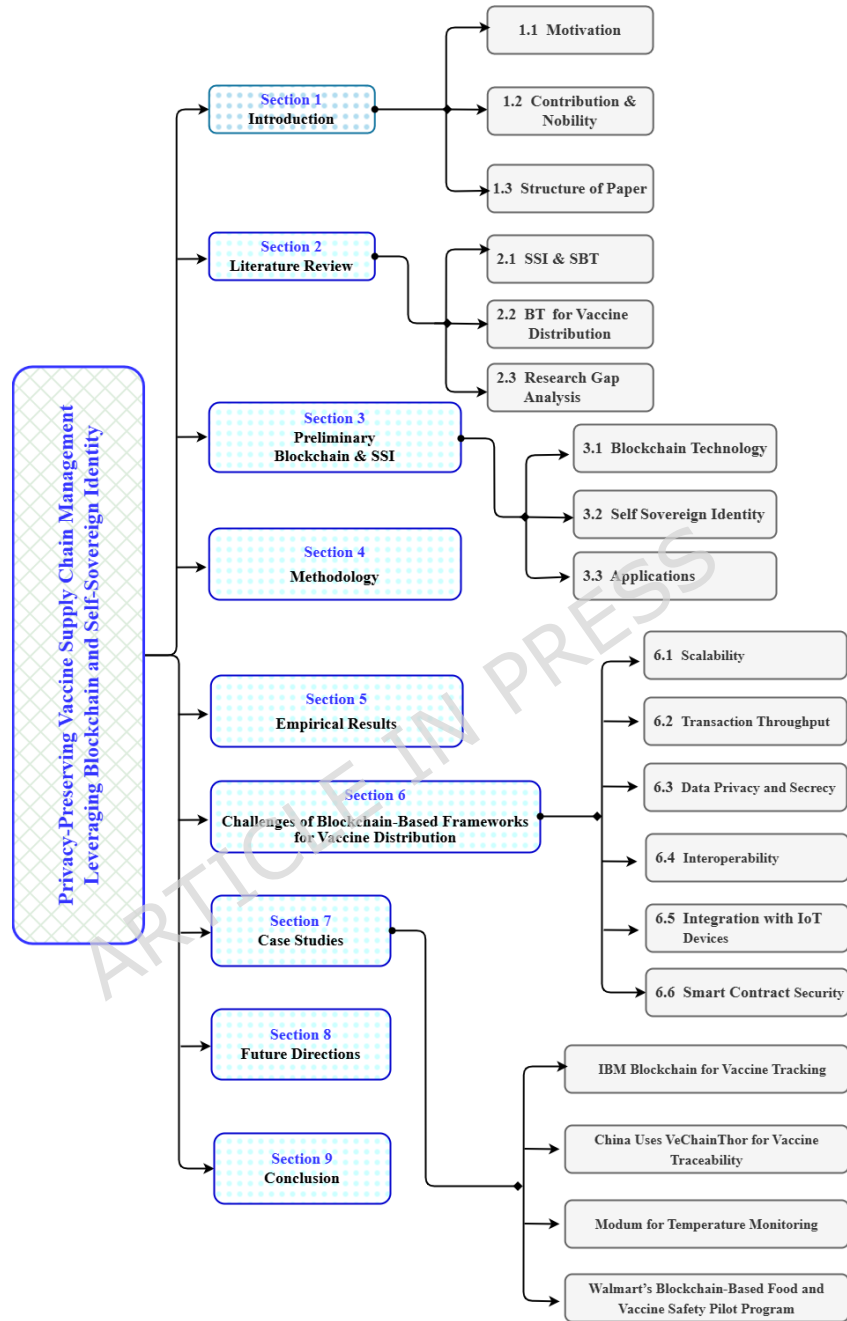


Fig. 3. Detailed organization of the content flow used in the paper

2 Related Work

Blockchain has been an active research area, with various applications leveraging SSI to preserve privacy. However, ensuring these models generalize well across diverse applications remains a significant challenge. This section provides an existing review and state-of-the-art (SOTA) discussion of BT, SSI, and hybrid mechanisms in supply chain management. Recent studies have explored blockchain-based secure and privacy-aware architectures in distributed and resource-constrained environments. For example, An et al. proposed a secure blockchain-enabled framework for decentralized energy sharing, emphasizing efficiency and trust among distributed participants, while Dong et al. presented a privacy-aware blockchain architecture for task distribution in IoT-enabled systems [83, 84]. Fig. 4 illustrates the process of considering existing work to identify research gaps, issues, and challenges in the drug supply chain system, followed by tabular analysis. We reviewed the literature on significant databases, including Google, IEEE, Springer, Elsevier, and Delnet. The effect of SSI and blockchain on drug supply chain privacy preservation and transparent traceability is also explored.

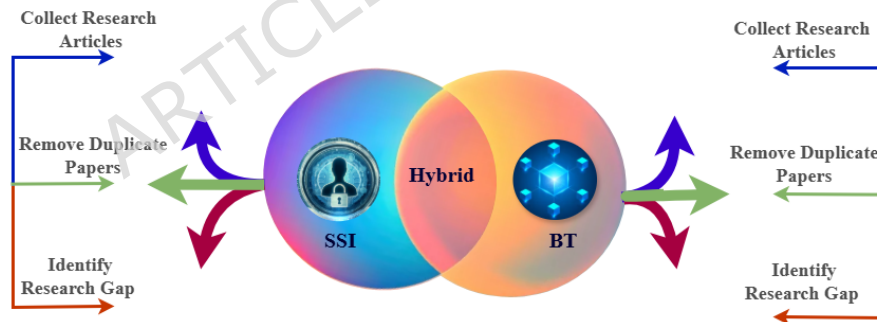


Fig. 4. Methodology applied for SOTA

2.1 SSI and BT

Guerar et al. [15] introduced SSI-MedRx, a blockchain- and SSI-based system designed to mitigate healthcare fraud, including phantom billing and opioid overprescription. The framework ensures privacy, authenticity, and cross-border interoperability by storing

verifiable credentials in patient-controlled wallets and leveraging blockchain to track prescriptions through MedRx tokens. Although primarily developed for healthcare, its use of decentralized identity, consent-based data sharing, and immutable records provides valuable insights for improving traceability and trust in pharmaceutical supply chains.

Similarly, a blockchain-based agri-food traceability system integrating IoT, IPFS, and SSI is presented in [16] to ensure transparency, data integrity, and security. While focused on food logistics, the adoption of verifiable credentials and decentralized storage is directly transferable to pharmaceutical traceability use cases and aligns conceptually with SSI-MedRx [15]. The work in [17] proposes a blockchain-IPFS framework to enhance traceability and safety in pharmaceutical supply chains, effectively addressing counterfeiting and data integrity through smart contracts. However, the absence of decentralized identity mechanisms and cross-border compliance limits its applicability in global contexts.

A comprehensive analysis of blockchain-based identity management and SSI is provided in [18], where privacy, identity challenges, and cost implications are systematically examined. In addition, decentralized anonymity and identity management systems are reviewed in [19], highlighting key research gaps, trade-offs, and constraints relevant to secure and scalable supply chain deployments.

2.2 BT for Vaccine Distribution

This sub-section reviews blockchain-based (BT) solutions for drug and vaccine supply chains that informed COVID-19 vaccination efforts. Blockchain has been widely adopted to ensure data integrity and immutability in beneficiary registration, thereby mitigating risks such as identity theft and impersonation [20]. Smart contracts have been employed to enforce safe handling conditions during vaccine distribution and to support structured reporting of adverse vaccination events, linking beneficiaries with administered doses [21]. Several prototype implementations using Ethereum-based test networks demonstrate the feasibility of blockchain-enabled vaccine traceability and auditability.

To improve traceability and distribution efficiency, platforms such as StaTwig leverage blockchain to connect vaccine manufacturers and consumers while addressing cold-chain degradation through smart contract enforcement [22]. Similarly, blockchain-based frameworks for tracking COVID-19 vaccination vials focus on secure

manufacturing-to-delivery monitoring, incorporating mechanisms such as proof of delivery and feedback logging to enhance transparency and accountability [23]. Enterprise-grade blockchain solutions using Hyperledger Fabric and Hyperledger Besu have also been explored to meet pharmaceutical traceability requirements, including privacy, authorization, authentication, and scalability [24]. Beyond traceability, some studies integrate blockchain with data analytics and machine learning to address challenges such as vaccine expiration, record fraud, and demand optimization [25]. Ethereum-based approaches combining smart contracts with off-chain storage have further demonstrated cost-effective and secure management of large-scale vaccination data, supported by performance and security evaluations [26]. Prototype implementations confirm the practicality of blockchain-based vaccination tracking while acknowledging the need for richer data models and real-world organizational validation [27].

Prior work has demonstrated that blockchain can serve as a coordination and trust layer in distributed learning and edge-based systems, although these solutions often lack identity-centric privacy guarantees and domain-specific governance mechanisms [86]. More recent works propose permissioned blockchain frameworks to enhance pharmaceutical traceability and counterfeit prevention, emphasizing secure access control and real-time visibility [28]. Blockchain-IoT integrations using RFID sensors extend these capabilities by linking physical drug movement with immutable digital records, although identity verification and patient-centric access control remain largely unaddressed [29]. Comprehensive reviews further highlight that, despite improvements in transparency and trust, most existing solutions lack integration with decentralized identity mechanisms such as SSI and fail to support interoperable, cross-border drug logistics [30]. Similar limitations are observed in blockchain-based healthcare supply chain frameworks that focus on smart contract-driven lifecycle monitoring without addressing decentralized identity or global regulatory interoperability [31].

2.3 Research Gap Analysis

Blockchain technology has gained much attention recently since it provides decentralized ways to create and manage significant data and associated transactions [38]. Before deployment in vaccine supply chain management, scalability, privacy-preserving, transparency, and traceability issues must be addressed. Data on vaccine identification and vaccination rates are essential for a nation to produce and distribute vaccines to achieve the targeted

immunization levels. The supply chain framework must be traceable and preserve privacy, especially for the immunization, temperature, batch number, and expiration date [39]. So, a digital identity for a product that is widely acceptable in the world is required. Blockchain technology with SSI is needed for the vaccine supply chain system [40]. To address the research gap analysis, we show a Table 2 based on cutting-edge technology used, use case, relevance to drug traceability, and limitations.

Table 2 Comprehensive Comparison of Cutting edge Technologies with its limitations

S. No.	Authors & Year	Technology Used & Use Case	Relevance to Drug/Vaccine Traceability	Limitations (Layer-Specific)
1	Guerar et al., 2025 [15]	Blockchain, SSI - Healthcare fraud prevention (SSI-MedRx)	Provides privacy-preserving, verifiable credential-based tracking applicable to pharmaceutical supply chains	Identity layer limited to healthcare context; no integration with physical logistics or cold-chain monitoring
2	Khoualdi et al., 2025 [16]	Blockchain, IoT, IPFS, SSI - Agri-food traceability	Demonstrates decentralized identity and immutable traceability transferable to drug logistics	Governance layer lacks pharmaceutical regulatory alignment; no support for drug-specific compliance workflows
3	Hasan et al., 2025 [17]	Blockchain, IPFS, Smart Contracts - Supply chain security	Addresses counterfeiting and data integrity in pharmaceutical supply chains	Identity layer lacks SSI-based authentication; interoperability limitations for cross-border deployments
4	Uddin, Mueen 202 [28]	Hyperledger Fabric - Pharmaceutical drug traceability	Ensures secure, permissioned traceability across pharmaceutical stakeholders	Identity layer relies on centralized MSP; limited interoperability with external

S. No.	Authors & Year	Technology Used & Use Case	Relevance to Drug/Vaccine Traceability	Limitations (Layer-Specific)
5	Ahmed et al., 2024 [29]	Blockchain, IoT, RFID Real-time drug tracking	Links physical drug movement with immutable digital records	regulatory systems Absence of decentralized identity; no patient-centric or role-based access control
6	Famous et al., 2025 [30]	Blockchain, Polymorphic Encryption, Cloud Drug supply chain management	Enhances transparency and confidentiality in drug logistics	Identity layer lacks SSI; integration complexity between encryption layer and blockchain
7	Chauhan et al., 2021 [32]	Ethereum, DApps Vaccine traceability	Demonstrates transparency and counterfeit prevention	Consensus layer scalability constraints; smart contract upgrade complexity
8	Bandhu et al., 2023 [20]	Ethereum, Smart Contracts Vaccine supply chain automation	Improves traceability and reduces fraud in vaccine distribution	Governance layer unclear; interoperability challenges with healthcare IT systems
9	Gao et al., 2023 [25]	Blockchain, Vaccine analytics and coordination	AI Supports predictive analytics for vaccine distribution	Data privacy risks at analytics layer; identity management not decentralized
10	Kumar et al., 2023 [33]	Hyperledger Fabric/Besu Enterprise drug traceability	Enhances trust and traceability in enterprise pharmaceutical networks	Limited scalability under high transaction load; consortium governance dependency
11	Hasan et al., 2019	Ethereum, Shipment	IoT Improves and shipment	Tight coupling between IoT and

S. No.	Authors & Year	Technology Used & Use Case	Relevance to Drug/Vaccine Traceability	Limitations (Layer-Specific)
	[34]	cold-chain monitoring	efficiency and monitoring accuracy	smart contracts; no SSI-based source authentication
12	Musamih et al., 2020 [35]	Blockchain Vaccination process support	Enhances transparency and tracking of vaccination events	High operational cost at consensus layer; limited scalability
13	Nabil et al., 2021 [36]	Blockchain Digital vaccine passport	- Ensures vaccination record integrity	Privacy concerns at identity layer; centralized identity assumptions
14	Alam et al., 2022 [37]	IF-DEMATEL, Blockchain Vaccine supply chain challenges	Identifies critical challenges and optimization priorities	Conceptual focus only; no identity layer or implementation framework

3. Preliminary Blockchain & SSI

Blockchain is a distributed ledger (DL) technology that allows for secure and transparent append-only transactions over a network [40]. It was initially launched as the core technology for the digital currency Bitcoin. However, it has subsequently found uses in other areas, including supply chain management, agriculture, education, finance, healthcare, energy sectors, and, more recently, vaccination delivery [41].

3.1 Blockchain Technology

Blockchain technology (BT) offers several properties that are highly relevant to vaccine supply chain management, including decentralization, transparency, traceability, immutability, security, and smart contract support [62]. These characteristics enhance the reliability and integrity of vaccine distribution processes by enabling tamper-resistant data sharing among distributed stakeholders. Blockchain-based zero-trust and permissioned governance models

have also been investigated to mitigate insider threats and ensure secure coordination among heterogeneous stakeholders in distributed systems [85]. The key features of blockchain technology are summarized in **Fig. 5** and briefly discussed below.

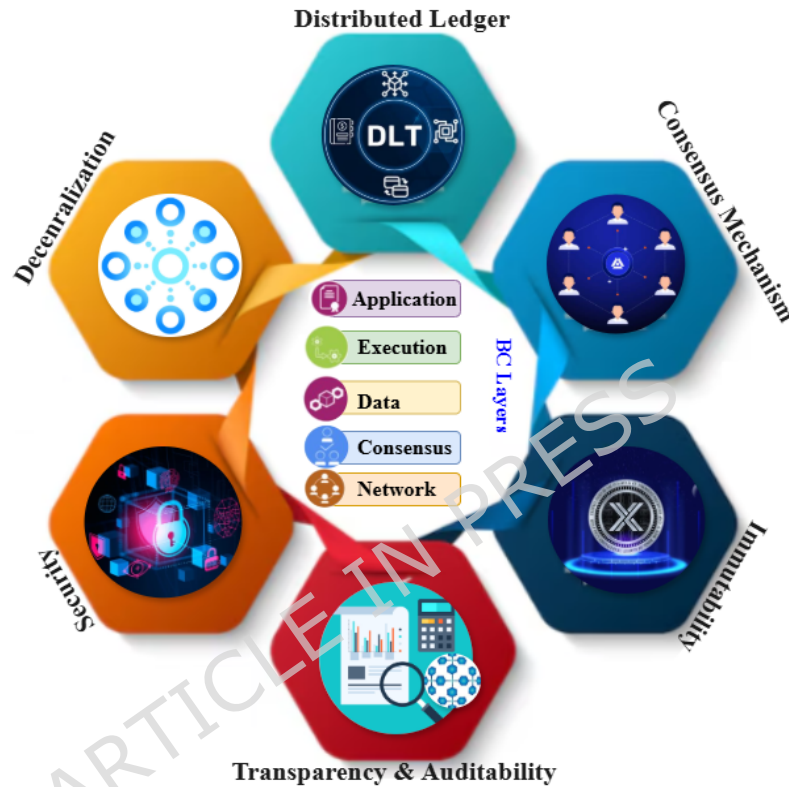


Fig. 5. Salient Features of BT

- **Decentralization:** Blockchain operates on a peer-to-peer architecture in which multiple nodes maintain synchronized copies of the ledger, eliminating dependence on a single controlling entity. This decentralized structure reduces single points of failure and improves resistance to tampering and censorship [42].
- **Distributed Ledger:** Transactions are recorded in a shared, append-only ledger, where each block is cryptographically linked to the previous one. This structure ensures consistency and data availability across all participating nodes [43].
- **Consensus Mechanism:** Blockchain networks rely on consensus algorithms to validate transactions and maintain

ledger integrity. Common mechanisms include Proof of Work (PoW), Proof of Stake (PoS), and Practical Byzantine Fault Tolerance (PBFT), each offering different trade-offs in terms of performance, scalability, and fault tolerance [44]. In many architectures, consensus is handled as a logical layer separate from application functionality [45].

- **Immutability:** Once validated and added to the ledger, blockchain transactions cannot be altered or removed. This immutability ensures long-term data integrity and is particularly valuable for applications such as vaccine distribution, where traceability and auditability are critical [46].
- **Transparency and Auditability:** Blockchain enables authorized participants to access the complete transaction history, from the genesis block onward. This transparency facilitates auditing, accountability, and trust among supply chain stakeholders [47].
- **Security:** Cryptographic techniques, including digital signatures and hashing, are used to protect transaction authenticity and data integrity. Combined with consensus mechanisms, these features make unauthorized modification or tampering computationally infeasible [48].

3.2 SSI

Self-Sovereign Identity (SSI) enables decentralized and privacy-preserving digital identity management through two core components: Decentralized Identifiers (DIDs) and Verifiable Credentials (VCs) [49]. Together, these components allow individuals and organizations to control identity data without reliance on centralized authorities. SSI enables authenticated participation without assuming trust in individual stakeholders, as authorization is enforced through verifiable credentials rather than centralized identity providers

- **DID:** A DID is a globally unique identifier that enables users to create, control, and share identity-related data in a decentralized manner. Standardized by the World Wide Web Consortium (W3C), DIDs uniquely identify both human and non-human entities and are associated with DID documents that specify cryptographic keys and service endpoints [50]. This structure enables secure identity resolution and management without centralized identity providers. [50].
- **Verifiable Credential (VC):** VCs are cryptographically secured digital credentials that represent claims about an

identity. An issuer generates a VC based on verified attributes and provides it to the holder, who retains full control over its storage and presentation [51]. During authentication, the holder selectively presents the credential to a verifier, who validates its authenticity and integrity without accessing unnecessary personal data.

BT's essence is to provide a decentralized and trustless system. In SSI, users have greater control over their data, assets, and interactions. The idea of sovereignty is illustrated in Fig. 6 as it emerges from the fundamental ideas of blockchain:

- **Autonomy:** Users have control over their own resources and actions in a BT [52]. Peer-to-peer communications and transaction control are attainable without the need for a central authority.
- **Decentralized:** The decentralized feature of blockchain is tightly linked with sovereignty. A decentralized network lacks a single point of failure or control [53]. The elimination of centralized authority or middlemen in identity management is referred to as the decentralization element of SSI.
- **Trust:** SSI enables trust through cryptographic mechanisms, decentralized networks, and verifiable credentials [54]. The blockchain's transparency and immutability guarantee that once a transaction is recorded, it cannot be changed, giving user's confidence in the system's integrity.
- **Transparency:** Transparency in SSI refers to how the identity framework ensures verifiable and auditable identity transactions with privacy-preserving or anonymous recorded transactions. Users can independently verify transactions and ensure the integrity of the supply chain management framework.



Fig. 6. Salient Features of SSI

3.3 Applications of Blockchain in Various Industries

BT has found applications across various fields, such as financial services, supply chain management, real estate, voting and governance, intellectual property rights, healthcare and many more [83]. Fig. 7 explores the categories overview of the significant applications of blockchain technology:

- **Financial Services:** Blockchain gained prominence through cryptocurrencies such as Bitcoin and has since been applied to cross-border payments, remittances, trade finance, and asset tokenization. Platforms such as Ripple and Stellar enable faster and more transparent transactions, while smart contracts automate and secure financial agreements [55, 80].
- **Supply Chain Management (SCM):** In supply chains, blockchain enhances transparency, traceability, and operational efficiency by providing end-to-end visibility of goods movement from production to delivery. Industry deployments, including IBM Food Trust and Walmart, demonstrate its effectiveness in provenance tracking, counterfeit prevention, and inventory management [56].

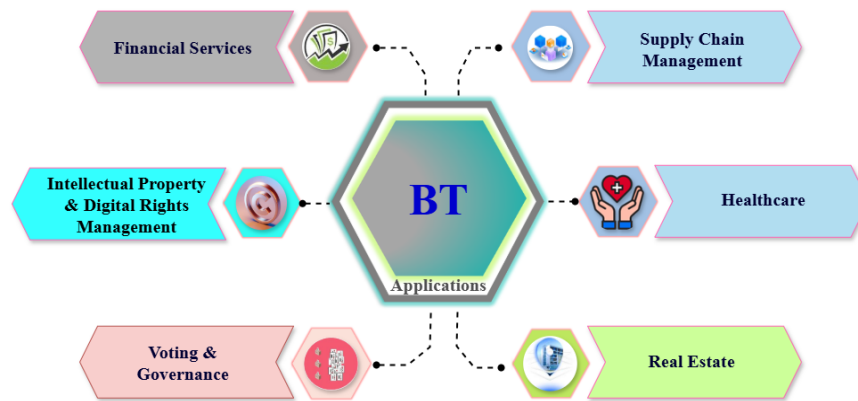


Fig. 7. Applications of Blockchain Technology

- **Real Estate:** Blockchain streamlines property transactions by enabling secure and transparent recording of ownership, titles, and contracts. Solutions such as Propy and Ubitquity reduce fraud and reliance on intermediaries, improving efficiency in real estate registration and management [57].
- **Voting and Governance:** Blockchain-based voting systems aim to ensure tamper-resistant, transparent, and auditable elections while preserving voter privacy. Platforms such as Voatz and Follow My Vote illustrate the potential of blockchain to improve trust and participation in democratic processes [58, 81].
- **Intellectual Property and Digital Rights Management:** Blockchain supports secure management of intellectual property by enabling timestamped ownership records and transparent licensing. Platforms such as Mediachain and Ujo Music allow creators to protect and monetize digital assets without relying on traditional intermediaries [59].
- **Healthcare:** Blockchain has significant potential to improve healthcare data interoperability, security, and patient-centric data control. Blockchain-enabled electronic health records (EHRs) support secure data access and sharing among healthcare providers, while also enabling pharmaceutical companies to track drug authenticity and provenance [60,61]. Early implementations, such as Medical Chain and Health Wizz, demonstrate patient empowerment through decentralized health data management.

Within healthcare ecosystems, blockchain applications extend to pharmaceutical companies, healthcare providers, insurance systems, government services, and consumers. These applications focus on securing sensitive data, reducing fraud,

automating administrative processes, and improving transparency and trust across stakeholders [62].

Finally, in addition to banking, blockchain technology has shown itself to be flexible and has the potential to disrupt a number of other industries, including supply chain management, healthcare, real estate, voting and governance, solar energy, electricity grid distribution, education, and intellectual property. These earlier blockchain implementations offer insightful information and lessons learned that can guide the creation and application of blockchain solutions for the COVID-19 vaccine distribution program and other pressing societal issues.

4 Methodology

This section presents a structured Vaccine Supply Chain Management System. As illustrated in Fig. 8, the methodology is divided into seven key phases: Vaccine Manufacturer, Air Transport, Primary Store, Refrigerated/Insulated Van, Sub-Center/Hospital/Health Center, Local/District Vaccine Store, and Patient/Person.

Phase One (Vaccine Manufacturer) initiates with data preparation as batch number, manufacturing date, expiry date, and required temperature, ensuring that it is batch-wise, day-wise, order-wise, supplier-wise, city-wise, state-wise, and country-wise. This stage involves data that can be hashed or QR code-wise hashed and minted in blockchain.

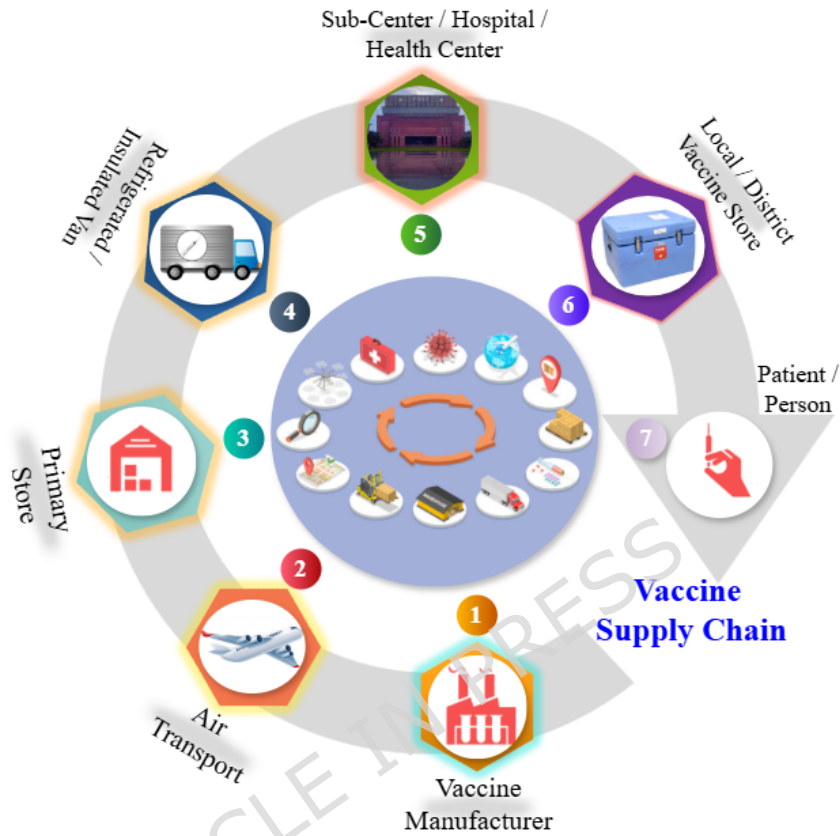


Fig. 8. Vaccine Supply Chain Management System

The proposed vaccine supply chain management assumes phase two (air transport) due to limited expiration time, temperature, and cost. During COVID-19, transportation was also an issue so that it may be shipped *via* other transportation to major hubs (nationwide) or state warehouses. At the time of shipping, monitoring for temperature is also possible due to minted blockchain entities as per the prescription of the vaccine about temperature and expiry of the vaccine.

Phase Three (Primary Store) preserves the vaccines in cold storage per prescribed norms. We have recommended IoT-enabled cold storage for vaccine preservation at a prescribed temperature [13], [63]. If the temperature is low, fluctuating, or high, it will be informed and stored in the cloud. This phase also includes generalization analysis, where the individual vaccine units are

preserved at the proper temperature and notified to the supplier, local hospitals, and patients are notified for transparency.

Phase Four (Refrigerated/Insulated Van) focuses on transporting from Cold chain [72] to sub-centers/ hospitals/ health centers to prevent spoilage.

The IoT and GPS-enabled van monitors real-time locations and conditions such as temperature. The `MonitorStorage(...)` (Algorithm 1) alerts if conditions deviate and mint the value in the blockchain.

Phase five (Sub-Center / Hospital / Health Center) and Phase six (Local / District Vaccine Store) are the local-level vaccine supply process with the preservation of all advisories and the handling timings with a temperature of the vaccine also minted in the blockchain. If the sub-center finds any discrepancies, then all storage devices of the vaccine and the damaged sample are stored in the blockchain to remove the duplicity and reuse the sample or reuse the storage unit.

Finally, Stage seven (Patient/ person) connects to the patient's digital health records and is stored in the blockchain for the future. Suppose any complications arise from the person/patient. In that case, data may be used by the insurance company and court as evidence due to immutability, and the proposed system can also identify the exact location of damages or faults.

We also explore the algorithms (Algorithm 1 & Algorithm 2) and workflow process for a more understandable, precise, and realistic phase-wise detailed explanation of the proposed Privacy-Preserving Vaccine Supply Chain Management Leveraging Blockchain and SSI. SSI-Aware Precondition (Applicable to Algorithms 1 and 2) is before invoking any smart contract function, the caller must present a valid BBS+-signed verifiable credential (VC) that has been verified off-chain against trusted issuer public keys and checked for revocation using status lists. Authorization decisions are enforced by a policy engine based on VC-bound roles and attributes. Only upon successful verification and policy approval is the corresponding transaction submitted to the blockchain.

Fig 9 depicts the end-to-end data partitioning and control flow that couples SSI with the vaccine supply chain: in the SSI layer, the issuer grants a BBS+-signed VC for a batch, the manufacturer and custodian present selective-disclosure views to the verifier, and revocation is checked via the Status List; in the App layer, the gateway/adaptor invokes `registerBatch(b, hash(VC))` and policy services authorize actions using VC-bound roles; in the Blockchain layer, the VaccineChain smart contract records custody events and anchors, and the Event Log provides immutable auditability (e.g., *BatchRegistered*, *Transferred*, *Alert*, *Disposed*); finally, the Off-Chain Storage keeps sensitive payloads—telemetry and

cryptographic evidence—encrypted, with only hash pointers (CIDs/URIs) committed on-chain. The arrows highlight privacy-preserving interactions: issuance → presentation/verification, policy-gated transferBatch(..., anchor), periodic monitoring (“write readings”), and verifier lookups (“fetch/verify proofs”). This layered design achieves data minimization (events + anchors on-chain), authenticity (VC signatures), revocability (status checks), and auditability (event logs), while keeping PII and raw operational data off the ledger.

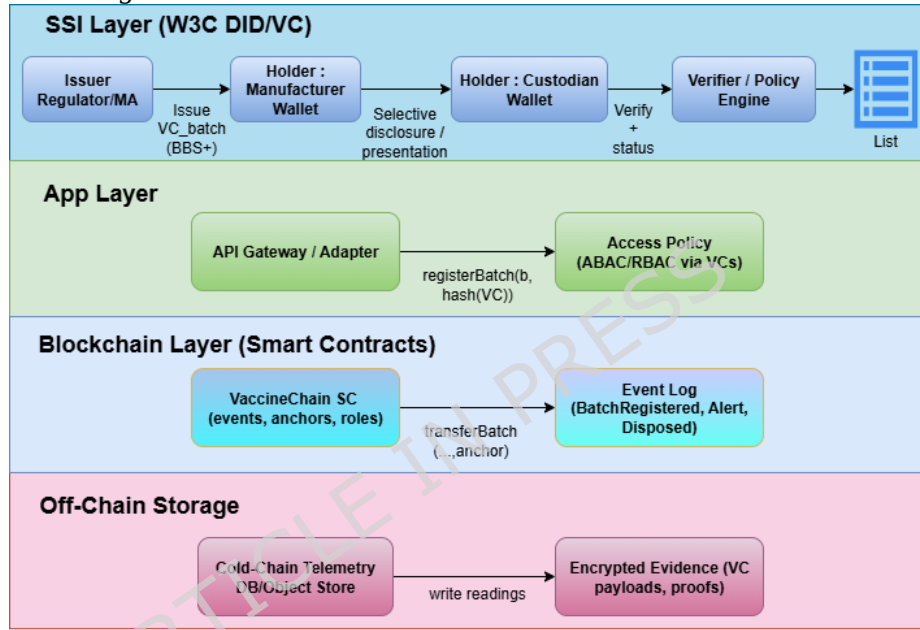


Figure 9: SSI-blockchain architecture for vaccine custody.

The algorithms use keywords such as *msg.sender* to identify the current caller (e.g., manufacturer, approver). *REQUIRE* ensures that a condition is met; otherwise, the transaction fails. *EMIT* broadcasts an event (e.g., for logs or monitoring). *Ownership*, *VaccineStatus*, *VaccineDetails*, and *StorageConditions* are state variables.

Algorithm 1 ensures secure and traceable operations such as registering, approving, rejecting, transferring, and monitoring vaccine batches using smart contract functions. All function invocations assume prior off-chain verification of role-specific verifiable credentials (e.g., manufacturer, approver, distributor) using pairwise decentralized identifiers (DIDs) and selective disclosure. It has four steps; the first step is the registration process using the *function* RegisterVaccineBatch(VaccineBatchID, VaccineDetails, Manufacturer). The step is responsible for

registering a new vaccine batch and storing the values of the new batch. The function is designed for the manufacturer's unit. The second step of the algorithm is to approve the registered batch of vaccines. The pending status is completed in the approval state by the function `ApproveVaccineBatch(VaccineBatchID, Approver)`. The third step is rejecting a batch if it fails quality or safety checks. The function is to handle the rejection process by `RejectVaccineBatch(VaccineBatchID, Approver, Reason)`. The proposed Blockchain supply chain uniqueness addresses the used and rejected vaccine units to stop duplicate vaccines and reuse the vaccine units. The fourth step is shipping the vaccine batch to another stakeholder, a national-level distributor, a hospital, centers, etc. The process uses `TransferVaccine(VaccineBatchID, To)` for the transit state after approval of the vaccine batches. The final fifth step for monitor storage conditions. The function `MonitorStorage(VaccineBatchID, ConditionData)` updates and monitors storage condition and also monitors in transit phase.

Although the proposed framework employs advanced SSI mechanisms such as BBS+-signed verifiable credentials, revocation status lists, pairwise decentralized identifiers (DIDs), and selective disclosure, these operations are intentionally performed off-chain. This design choice follows established SSI architectures, where credential issuance, presentation generation, signature verification, revocation checks, and zero-knowledge predicate evaluation are executed by SSI agents or middleware layers rather than smart contracts.

The blockchain layer is therefore used only to anchor the results of successful SSI verification, record custody events, and provide immutable auditability. Algorithms 1 and 2 abstract these interactions by assuming that any on-chain transaction invoking registration, transfer, administration, or disposal functions is preceded by successful off-chain SSI verification and policy evaluation.

Algorithm 1 Registration Phase with Vaccine Approval, Reject, Transfer, and Monitor Vaccine Batches

Input: Entries regarding vaccine batches approval, reject, transfer, and monitor

Output: Securely storage of vaccine batches status

```

1: Step 1: Registration Process
2: RegisterVaccineBatch(VaccineBatchID, VaccineDetails,
  Manufacturer)
3: begin
4:   if msg.sender  $\neq$  Manufacturer
5:   then REQUIRE;
6:     VaccineStatus[VaccineBatchID]  $\neq$  NULL
7:   then REQUIRE;
8:     VaccineBatchInfo[VaccineBatchID] =
VaccineDetails;
9:     VaccineStatus[VaccineBatchID] = Pending;
10:    VaccineRegistered(VaccineBatchID, Manufacturer,
NOW);
11:  end
12: end
13: Step 2: Approval of Vaccine Batches
14: ApproveVaccineBatch(VaccineBatchID, Approver)
15: begin
16:   if msg.sender  $\neq$  Approver
17:   then REQUIRE;
18:     if VaccineStatus[VaccineBatchID]  $\neq$  Pending
19:     then REQUIRE;
20:       VaccineStatus[VaccineBatchID] = Approved;
21:       VaccineApproved(VaccineBatchID, Approver,
NOW);
22:     end
23:   end
24: end
25: Step 3: Rejection Process of Vaccine Batches
26: RejectVaccineBatch(VaccineBatchID, Approver, Reason)
27: begin
28:   if msg.sender  $==$  Approver
29:   then REQUIRE;
30:     if VaccineStatus[VaccineBatchID]  $\neq$  Pending

```

```

31:         then REQUIRE;
32:         VaccineStatus[VaccineBatchID] = Rejected;
33:         VaccineRejected(VaccineBatchID, Approver,
Reason, NOW);
34:     end
35: end
36: end

37: Step 4: Shipping the vaccine batch to another
stakeholder may be
38:     national level distributor, hospital, centers, etc.
39:     TransferVaccine(VaccineBatchID, To)
40:     begin
41:         if VaccineStatus[VaccineBatchID] == Approved AND
42:         VaccineStatus[VaccineBatchID] == InTransit
43:         then REQUIRE;
44:         if msg.sender == Ownership[VaccineBatchID]
45:         then REQUIRE;
46:         Ownership[VaccineBatchID] = To;
47:         VaccineStatus[VaccineBatchID] = InTransit;
48:         VaccineTransferred(VaccineBatchID,
msg.sender, To, NOW);
49:         end
50:     end
51:     end

52: Step 5: Monitor storage conditions of Vaccine Batches
53:     MonitorStorage(VaccineBatchID, ConditionData)
54:     begin
55:         if VaccineStatus[VaccineBatchID] == InTransit
56:         then REQUIRE;
57:         if ConditionData != Standards
58:         then REQUIRE;
59:         EMIT StorageAlert(VaccineBatchID,
ConditionData, NOW);
60:         StorageConditions[VaccineBatchID] =
ConditionData
61:         end
62:     end
63:     end

```

4.1 SSI-Blockchain Interaction & Transaction Flow

The flow in Figure 10 binds each custody event to SSI while keeping payloads off-chain. The manufacturer receives a VC for a batch (lot, expiry, temp policy), then anchors only its hash via registerBatch. At each handoff, the holder presents a selective-disclosure VC to a verifier; on success, the holder commits transferBatch with a hash of the presentation/proof. Pairwise DIDs and canonicalized presentations reduce linkability; revocation lists are checked before transfers. During cold-chain monitoring, monitorStorage emits StorageOK/StorageAlert events, while raw telemetry stays encrypted off-chain. When a dose is given, the clinic calls administerDose with a hashed patient DID and anchors evidence by hash; disposal closes the lifecycle with disposeBatch. Security follows from VC signatures + role-gated calls; integrity and non-repudiation from on-chain events and anchors; privacy from selective disclosure and off-chain storage; replay is mitigated by receiver-bound presentations with nonces/timestamps. Verification is off-chain, so on-chain writes are small and final quickly on a permissioned chain; optional ZKP predicates add only verifier CPU

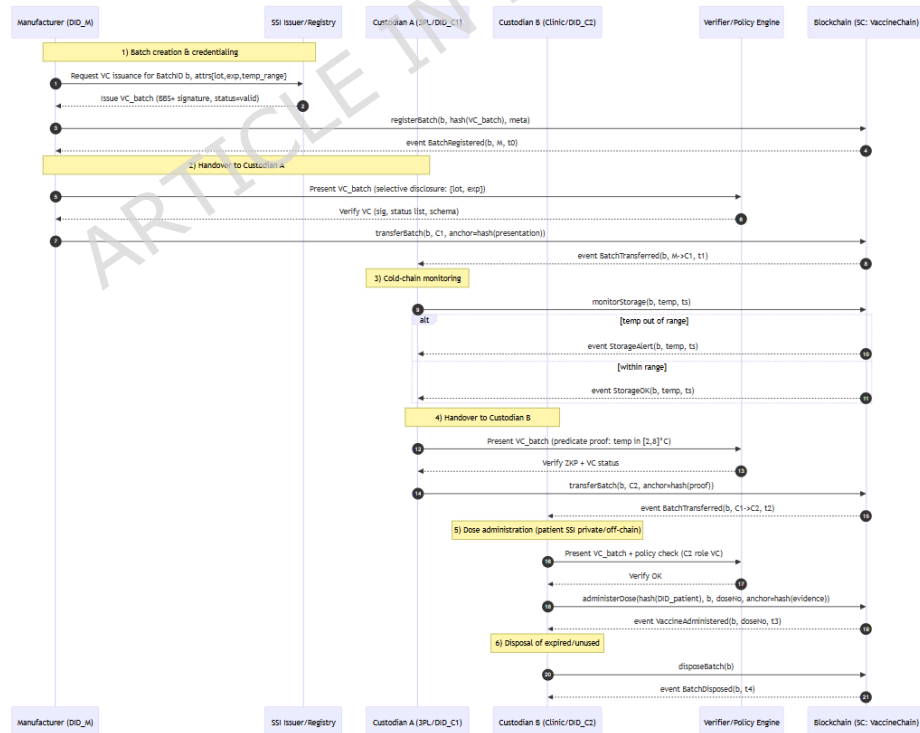


Figure 10: Sequence Diagram SSI-Blockchain Interaction

Algorithm 2 uses a blockchain-based vaccine supply chain management system. It defines three smart contract functions to confirm that a receiver has received a vaccine batch, mint a vaccine dose administered to a patient, and dispose of expired or unusable vaccine batches.

In Algorithm 2, the first step is to confirm that a receiver process for a health center or hospital has received a vaccine batch. `ConfirmVaccineReceipt(VaccineBatchID, Receiver)` is used for delivered status with a timestamp for transparency and traceability. The second step is minting the vaccine dose information administered to a patient. `AdministerVaccine(PatientID, VaccineBatchID, DoseNumber)` is used to store patient information, batch numbers, and ID information in the vaccine supply chain system. Finally, the last step is to dispose of expired or unusable vaccine batches. We have stored used vaccines and disposed of vaccine container information to stop reuse and expired or unusable information to reduce misuse or in favor of patient health. For Algorithm 2, zero-knowledge proof (ZKP) predicates such as authorization without revealing identity attributes are verified off-chain before committing transaction hashes or anchors on-chain.

Algorithm 2 Vaccine Received and Disposal Phase

Input: Entries regarding vaccine batches confirmation and dispose vaccine

Output: Securely traceable dispose of expired or unusable vaccine batches.

```

1: Step 1: Received a vaccine batch process
2:   Confirm VaccineReceipt(VaccineBatchID, Receiver)
3:   begin
4:     if msg.sender ≠ Manufacturer
5:       then REQUIRE;
6:         VaccineStatus[VaccineBatchID] != NULL
7:       then REQUIRE;
8:         VaccineStatus[VaccineBatchID] ≠ InTransit;
9:         VaccineStatus[VaccineBatchID] = Delivered;
10:        Ownership[VaccineBatchID] = Receiver;
11:        VaccineReceived(VaccineBatchID, Receiver, NOW);
12:     end
13:   end

```

```

14:   Step 2: Minted in blockchain information
regarding vaccine dose has
15:   been administered to a patient.
16:   Administer Vaccine(PatientID, VaccineBatchID,
DoseNumber)
17:   begin
18:   | if VaccineStatus[VaccineBatchID]  $\neq$  Delivered
19:   |   then REQUIRE;
20:   |   | if VaccineStatus[VaccineBatchID]  $\neq$  Pending
21:   |   |   then REQUIRE;
22:   |   |   VaccineAdministered(PatientID, VaccineBatchID,
23:   |   |   |   DoseNumber, NOW);
24:   |   |   end
25:   |   end
26:   end
27: Step 3: Dispose of expired or unusable vaccine batches
28: DisposeVaccine(VaccineBatchID, Disposer)
29: begin
30: | if msg.sender  $\neq$  Disposer
31: |   then REQUIRE;
32: |   VaccineStatus[VaccineBatchID] = Disposed;
33: |   VaccineDisposed(VaccineBatchID, Disposer, NOW)
34: |   end
35: end

```

4.2 Threat Model and Trust Assumptions

The proposed blockchain-SSI-enabled vaccine supply chain framework is designed to operate in an adversarial environment involving multiple independent stakeholders. The considered threat model includes counterfeit manufacturers attempting to introduce fake or expired vaccine batches, malicious or negligent distributors that may alter custody records or mishandle vaccines, compromised IoT sensors that could report falsified temperature or location data, and dishonest clinics or vaccination centers that may misreport vaccine usage or administration events.

The system mitigates these threats through a combination of blockchain immutability and SSI-based identity assurance. Blockchain ensures that once custody events, temperature alerts, or disposal records are anchored on-chain, they cannot be altered retroactively. SSI mechanisms enforce authenticated participation by requiring stakeholders to present verifiable

credentials (VCs) bound to decentralized identifiers (DIDs), thereby preventing unauthorized entities from interacting with the system. Selective disclosure further limits information exposure while preserving verifiability.

The trust assumptions of the framework are intentionally minimized. Credential issuers such as government health authorities, regulators, and certified manufacturers are assumed to be honest and compliant with regulatory requirements when issuing VCs. Verifiers, including logistics providers and healthcare facilities, are assumed to correctly validate credential authenticity and revocation status during interactions. The blockchain operates under a permissioned trust model, where a majority of validator nodes operated by authorized institutions are assumed to behave honestly, ensuring consensus integrity without relying on a single trusted entity.

Importantly, the framework does not assume trust in individual distributors, clinics, or IoT devices. Instead, misbehaviour is detected through cross-verification of credentials, immutable audit trails, and anomaly detection in off-chain monitoring data. This design limits the impact of compromised actors while maintaining traceability, accountability, and privacy across the vaccine supply chain.

4.3 Governance and Trust Assumptions

The proposed vaccine supply chain framework operates under a permissioned blockchain governance model, reflecting the regulatory nature of pharmaceutical and vaccination ecosystems. Blockchain validator nodes are assumed to be operated by trusted institutional stakeholders, such as government health authorities, national regulators, vaccine manufacturers, and authorized distributors. These entities collectively participate in consensus to validate transactions and maintain ledger integrity, ensuring accountability while avoiding reliance on a single centralized authority.

Identity issuance within the system follows a Self-Sovereign Identity (SSI) trust model. Trusted authorities such as regulatory agencies, vaccine manufacturers, and accredited healthcare organizations act as credential issuers, generating verifiable credentials (VCs) for supply chain participants. These credentials certify roles and permissions (e.g., manufacturer, distributor, healthcare provider) and are bound to decentralized identifiers

(DIDs) controlled by the respective entities. Verifiers, including logistics providers and vaccination centers, validate credentials during interactions without accessing unnecessary identity attributes.

The trust assumptions of the proposed system are therefore limited to: (i) honest operation of credential issuers following regulatory compliance, and (ii) majority-honest behaviour among permissioned blockchain validators. No single entity is trusted with full control over identity, data, or transaction validation, thereby reducing centralized trust dependencies while preserving regulatory oversight

5 Empirical Results

This section presents the proposed system's implementation outcomes, focusing on the effectiveness of the Privacy-Preserving Vaccine Supply Chain Management Leveraging Blockchain and Self-Sovereign Identity framework. The results fulfill several critical performance and practical deployment gaps in the blockchain-SSI-based vaccine supply chain framework. Lack of empirical delay benchmarks by Offering concrete hash delay values across varied message sizes for real-time feasibility analysis. Uncertainty in resource consumption using the Ethereum blockchain provides gas usage and energy trends to evaluate cost-efficiency and sustainability. Scalability validation analyzes the performance metrics scale with increasing transactions, aiding deployment planning. System optimization planning is attained by supporting hardware selection and architectural decisions by linking performance with machine specs. Energy awareness in healthcare IT Enables energy analysis per transaction, which is helpful for eco-conscious medical infrastructure.

All experiments ran on Ubuntu 22.04 LTS with an Intel Core i7-12700H (12-core, 2.30 GHz), 16 GB DDR4, and a 512 GB SSD. The blockchain stack used Geth v1.13.7 and Solidity 0.8.x (optimizer on, 200 runs); MetaMask handled transaction signing, and Python (hashlib) produced content hashes for integrity checks. Our Ethereum baseline used a local Geth dev network (Clique PoA) configured with 1 s block time, 30 M block gas limit, EVM "London"+ features, and EIP-1559 gas accounting; Ganache was used only for isolated unit tests. In parallel, the permissioned deployment ran with 7 validators, PBFT-style finality, 200 ms block time, 15 M block-gas cap, and a 2 s view-change timeout. The workload came from a seeded synthetic generator (10k records; schema: {transaction_id, batch_id, manufacturer_id, receiver_id, temperature_C, expiry_date,

status, dose_count, timestamp}). Each experiment used 3 warm-up runs followed by 10 measured runs; we report mean $\pm$ 95% CI. Network conditions were emulated with 10-50 ms latency, 5 ms jitter, and 0-1% loss. Reported metrics include end-to-end custody latency, block commit time, TPS (1-s window), gas per transaction (network-independent “gas used”), and energy via Intel RAPL (or Watts Up Pro).

The dataset utilized in this study was synthetically generated to analyze cryptographic processing, blockchain transaction behavior, and energy-cost metrics within the context of a vaccine supply chain framework. Designed to reflect realistic operational scenarios, each record in the dataset represents a discrete transaction and includes attributes relevant to a blockchain-based traceability system. These attributes comprise essential identifiers such as *transaction_id*, *batch_id*, *manufacturer_id*, and *receiver_id*, facilitating source-to-destination tracking. Environmental and logistical parameters, including *temperature_C*, *expiry_date*, and *timestamp*, are incorporated to simulate cold chain monitoring and compliance with vaccine storage guidelines. Moreover, each record contains a status field (e.g., Approved, InTransit, Delivered, Rejected) to model different stages of vaccine lifecycle progression and a *dose_count* parameter indicating the number of doses associated with a given batch.

The performance evaluation presented in this study highlights the practical considerations and trade-offs between a public Ethereum-based blockchain implementation and a lightweight, Python-based customized blockchain framework tailored for secure and traceable healthcare applications. The core parameters analyzed include SHA-256 hash computation delay, transaction latency, energy consumption, and gas usage, measured against increasing message sizes ranging from 100 to 1000 records. Each message comprises 20 fields encoded using 30-bit data lengths, reflecting a realistic data model for vaccine batch records or electronic health events. Fig. 11 shows the Ethereum-based setup exhibited predictable increases in hash delays. It indicates that the SHA-256 delay grew linearly, from approximately 4800 microseconds for 100 messages to 48,000 microseconds for 1000 messages. However, for real evaluation, we have also analyzed the SHA-256 function using Python, and comparatively, the delays are not increasing linearly.

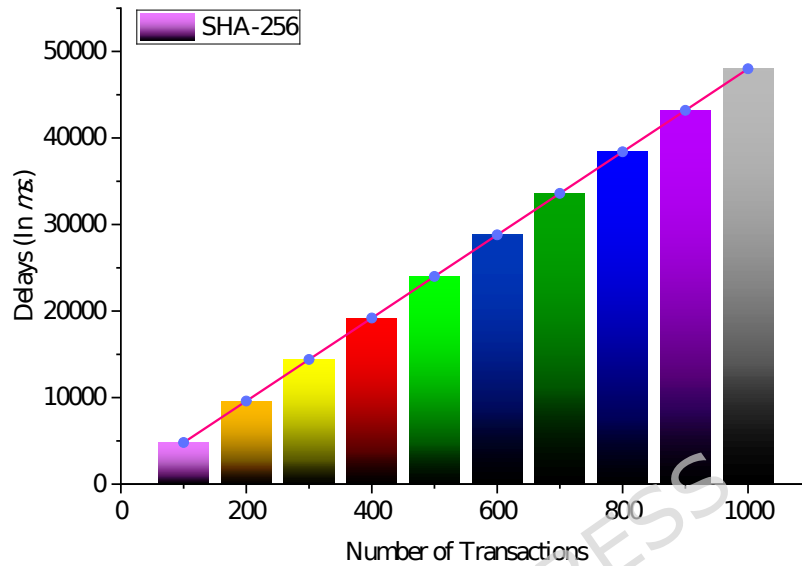


Fig. 11. Performance of Hash Function using Ethereum-based Setup

Fig. 12 shows the Python-based optimized blockchain setup exhibited predictable increases in hash delays. It indicates that the SHA-256 delay grew linearly, from approximately 1.5 microseconds for 100 messages to 15.3 microseconds for 1000 messages. It is significantly shallow in comparison to the Ethereum blockchain.

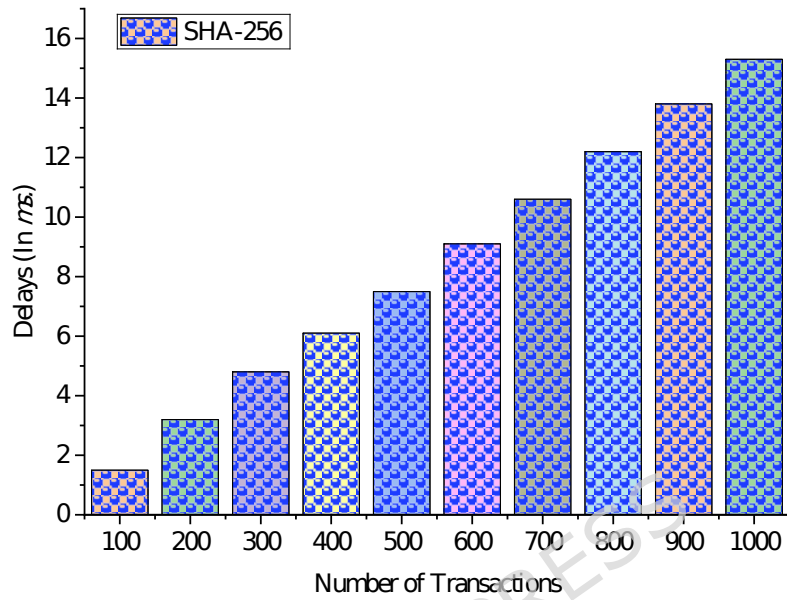


Fig. 12. Performance of Hash Function using Python-based Setup

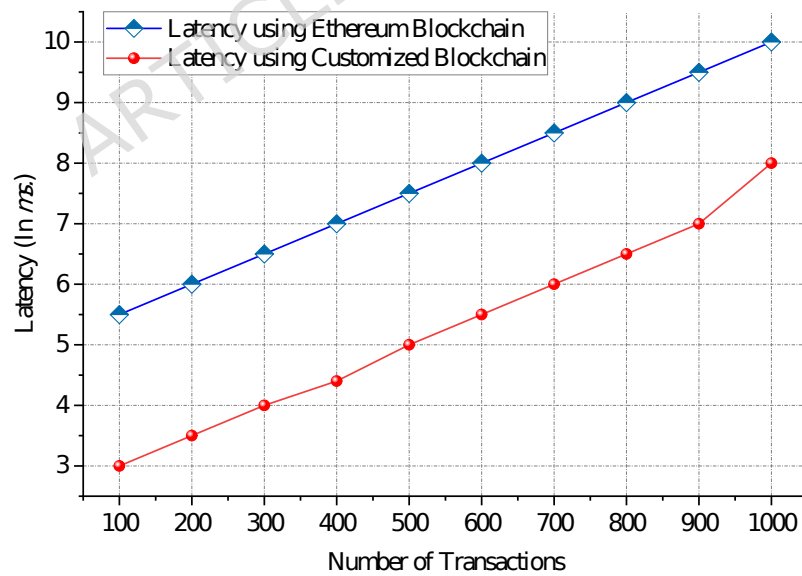


Fig. 13. Latency of the Proposed Supply Chain Management Framework

Fig. 13 shows the latency of blockchain & SSI-enabled supply chain management framework transactions. Ethereum transaction latency increased from 5.5 milliseconds to 10 milliseconds across the 100 to 1000 transactions exchanged. Addition to customized blockchain completes 3 to 8 milliseconds over the same range of exchanged transactions. By contrast, the Python-based customized blockchain exhibits markedly reduced latency and energy overhead. Implemented in a trusted, permissioned environment with simplified consensus (or file-based append-only mechanisms), it eliminates the need for expensive mining, gas metering, and complex transaction finality conditions.

Fig. 14 shows the real deployment latency of the proposed supply chain framework. It considers real-time delays that occur due to network congestion, internet speed, and internet glitches. We have found that at the highest transaction of up to 1000, the minimum and maximum latency delays are 380 *ms* and 450 *ms*, respectively. However, at lower transactions up upto 100 the latency minimum and maximum delays are 150 *ms* and 200 *ms* only.

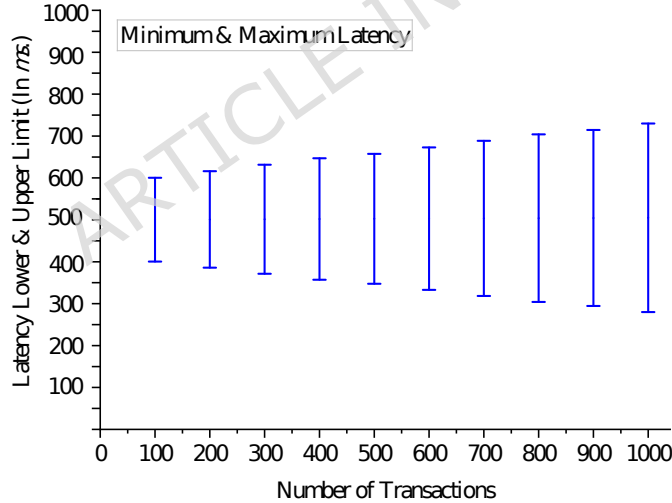


Fig. 14. Lower & Upper Limit Latency of the Proposed Supply Chain Management Framework

Fig. 15 shows processing power consumption, known as gas, varies with the number of transactions processed in a batch within

the blockchain & SSI-enabled supply chain management system for vaccines. It is also observed that the gas usage rises from 22,000 units to 31,000 units, reflecting Ethereum's fee structure over the 100 to 1000 transactions in the proposed framework. Moreover, this also highlights a gradual increase in gas usage with higher transaction volumes within the supply chain management system. However, these results assume idealized, isolated execution and do not account for the additional latency introduced in real-time deployments. When deployed on a public or testnet Ethereum network, the transaction life cycle involves multiple stages—network propagation, block confirmation, mempool queuing, and peer synchronization—all of which introduce cumulative latency that is not captured in pure computation-based simulations. Specifically, network propagation delays can add between 100 to 300 milliseconds, depending on node distribution and congestion, while block confirmation may introduce additional delays ranging from 5 to 15 seconds on testnets and up to 60 seconds on the mainnet. Mempool queuing and transaction prioritization may further delay execution, especially under high-throughput scenarios. For private or consortium-based blockchains utilizing Practical Byzantine Fault Tolerance (PBFT), synchronization overhead among validator nodes can add 10 to 50 milliseconds per peer, particularly when multiple institutions are participating in consensus.

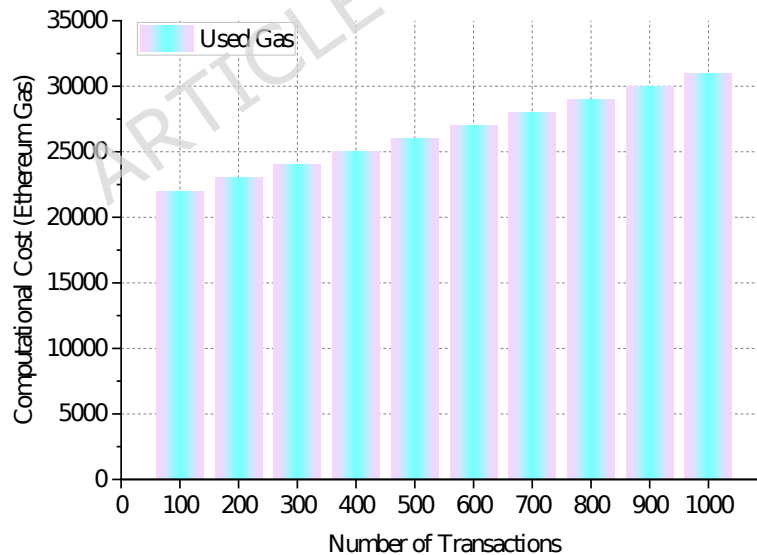


Fig. 15. Gas usage in Blockchain transactions of the Proposed Supply Chain Management Framework

Fig. 16 illustrates energy consumption in millijoules gradually increasing with an increasing number of transactions during a computational process. The energy consumption is observed to rise steadily from 2.2 mJ to 4.0 mJ over 100 to 1000 transactions. The result proves the efficacy of the proposed framework and also indicates a scalable and predictable energy model in terms of efficient computational scalability.

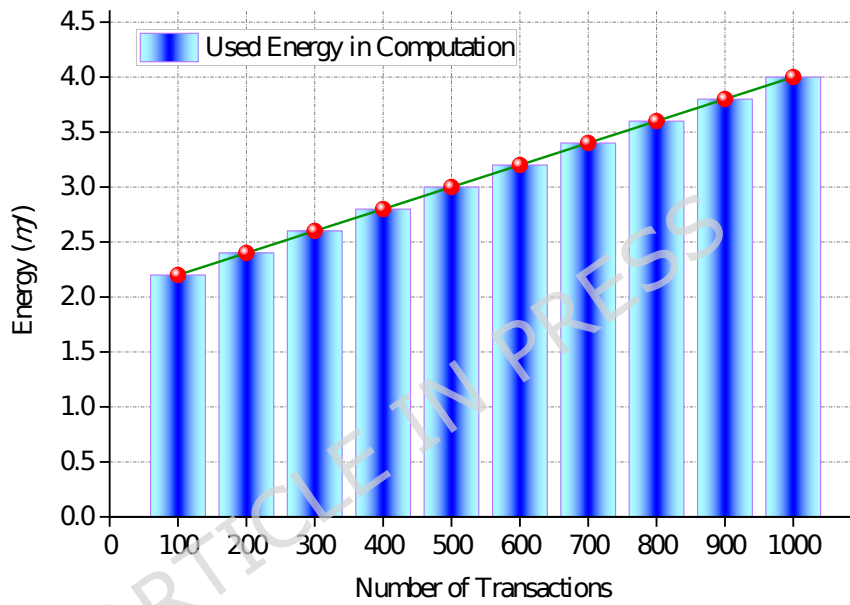


Fig. 16. Energy usage of the Proposed Supply Chain Management Framework

The time required to verify, complete, and add a new block to the blockchain ledger is known as the block commit time [74], [75]. It is the span of time that lies between a transaction being minted and the blockchain effectively committing its block. In the proposed system, SSI formation is offline, and the blockchain framework also provides transaction IDs. After identifying transactions appended in the blockchain, computation commit time is illustrated in Fig. 17. The committed time, or minting time, of transactions in blocks is 30 to 100 ms over 100 to 1000 transactions, respectively. These observations are particularly significant for applications where real-time responsiveness, data integrity, and system resource efficiency are paramount. While suitable for public verifiability and

open-access applications, the high overhead associated with Ethereum becomes a bottleneck in regulated environments requiring high transaction throughput, fast access times, and limited computational budgets. The customized blockchain model provides a viable alternative, enabling secure, low-latency, and energy-efficient data recording and verification for critical operations such as vaccine distribution monitoring, patient immunization logging, and cold chain compliance tracking.

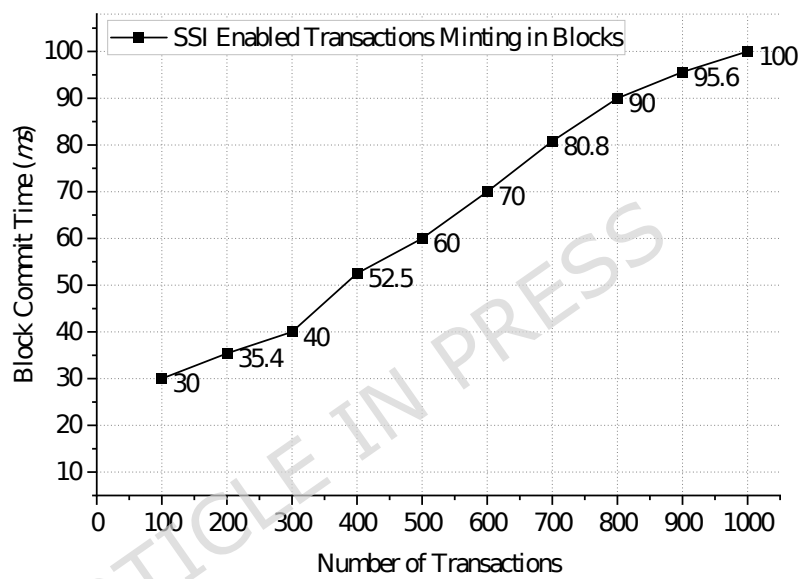


Fig. 17. Block Commit Time of Blockchain transactions

Omar et al. [76], Antal et al. [77], and Musamih et al. [78] also use the Ethereum blockchain for the supply chain system. The vaccine supply chain management framework was studied in [77] and [78]. Fig. 18 shows the comparative analysis of the proposed scheme with other existing schemes. The comparative analysis proves the efficacy of the proposed blockchain and SSI-enabled vaccine supply management framework. The [78] shows the significant performance of gas usage in health care registration only. However, it consumes 130,798 gas in the commitment process. At the same time, the proposed scheme consumes only 31,000 units of gas over 1,000 transactions.

These observations are particularly significant for applications where real-time responsiveness, data integrity, and system resource efficiency are paramount. While suitable for public verifiability and

open-access applications, the high overhead associated with Ethereum becomes a bottleneck in regulated environments requiring high transaction throughput, fast access times, and limited computational budgets.

The results validate the importance of evaluating blockchain systems not solely on cryptographic strength or decentralization but also on deployment context and operational performance under real-world conditions. It offers robust security and decentralized consensus; its practical utility in tightly constrained environments—such as point-of-care IoMT systems—is limited by its latency and resource demands. Conversely, the customized blockchain approach aligns well with healthcare’s performance and security requirements, balancing traceability, efficiency, and scalability.

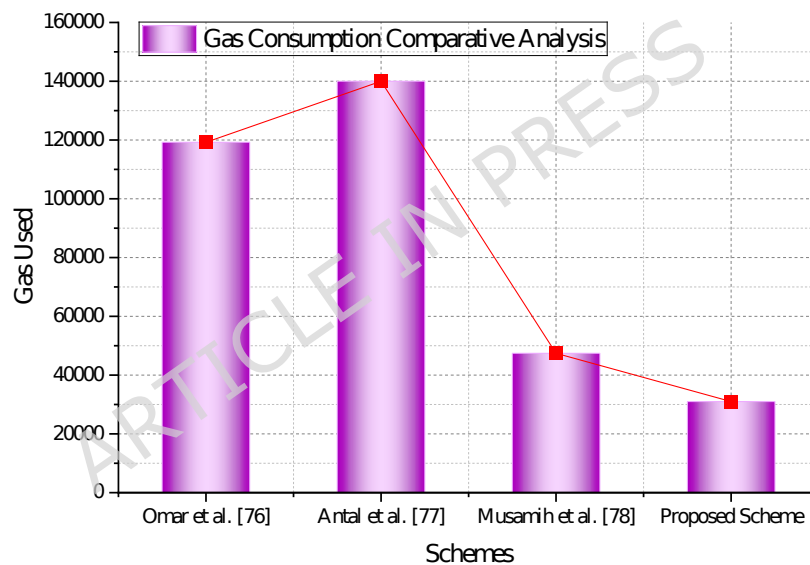


Fig. 18. Comparative Analysis with Existing Schemes

Table 3 summarizes how the architecture satisfies key GDPR/HIPAA obligations, data minimization by recording only events and hash anchors on-chain while keeping payloads off-chain; integrity and auditability via immutable custody events with anchor verification; revocability/rectification through VC status lists and DID/key rotation so access can be withdrawn and provenance updated; purpose limitation enforced by a policy engine that gates actions based on verified roles/VCs; and confidentiality by encrypting off-chain evidence and using selective disclosure in presentations.

Together, these controls align with GDPR's minimization, integrity, purpose limitation, and accountability principles and HIPAA's "minimum necessary" and access-control requirements while preserving end-to-end traceability (see Table 3).

Principle	Mechanism in our framework	Where enforced
Data minimization	On-chain stores only events + hashes; payloads off-chain	Contract events; off-chain store
Integrity & auditability	Immutable custody events; hash anchors	Event log; anchor verification
Revocability/ rectification	VC status lists; DID/ key rotation	SSI verifier; role revocation
Purpose limitation	Policy engine gates access by role/ VC	Verifier/ policy layer
Confidentiality (HIPAA)	Encrypted off-chain evidence; selective disclosure	Off-chain store; VC presentations

Table 3. GDPR/HIPAA compliance mapping: principles, mechanisms, and enforcement points.

The performance analysis underscores that application-specific constraints must guide architectural and deployment choices. The simulated and projected latency models demonstrate that lightweight blockchain solutions, tailored with domain-specific optimizations, are feasible and essential for achieving high-performance, secure, and trustworthy digital health infrastructures.

6 Challenges of Blockchain-Based Frameworks for Vaccine Distribution

Blockchain-based vaccination distribution methods encounter technical difficulties during implementation, such as scalability issues, where high transaction volumes can overwhelm the network. Interoperability between diverse systems is a challenge, complicating data integration. The six significant challenges are illustrated in Fig. 19. Ensuring data privacy while preserving transparency is complex, and integrating with existing infrastructure can be cumbersome. Additionally, high implementation costs and the technical complexity of blockchain technology can hinder practical deployment and usability. These factors have been discussed in detail below.



Fig. 19. Challenges of Vaccine Supply Chain Management System

6.1 Scalability

Developing blockchain solutions for vaccine distribution has many technological obstacles, including scalability. Blockchain networks, particularly open ones, might have trouble processing a lot of transactions rapidly. Scalability becomes a crucial concern when vaccine distribution requires a large number of transactions, such as tracking shipments, documenting temperature data, and confirming validity [64]. Adopting strategies to expand blockchain networks while preserving decentralization and performance is crucial to guarantee prompt and effective vaccine distribution.

6.2 Transaction Throughput

The quantity of transactions a blockchain network can handle in a given time is referred to as transaction throughput. High transaction throughput is essential for vaccine distribution systems to manage real-time tracking and certification of vaccine

shipments. However, because of consensus procedures, block size restrictions, and network congestion, many blockchain platforms have limitations in terms of transaction speed [65]. It is necessary to put solutions in place to get around these restrictions, such as sharding, off-chain scaling strategies, or high-throughput consensus algorithms.

6.3 Data Privacy and Secrecy

Although blockchain technology provides immutability and transparency, protecting data privacy and secrecy is a major concern, particularly in distributing vaccines where sensitive data, including patient and supply chain records, is involved. The intrinsic openness of blockchain technology may pose a challenge to privacy legislation like GDPR and HIPAA [66]. It is a complex technical task to implement privacy-protecting methods like encryption, permissioned blockchains, and zero-knowledge proofs while preserving transparency and audibility.

6.4 Interoperability

The distribution of vaccines frequently involves several parties, each of whom uses a different set of systems and platforms, including producers, distributors, healthcare providers, and regulatory authorities. For smooth data interchange and collaboration, these different systems and blockchain networks must be able to communicate with one another. However, due to variations in data formats, protocols, and governance frameworks, interoperability remains a technological problem [67]. Successful implementation requires the creation of middleware, standards, and protocols to enable interoperability between blockchain networks and traditional systems.

6.5 Integration with IoT Devices

IoT devices are essential for tracking shipments, monitoring vaccine storage conditions, and gathering real-time data for vaccine distribution networks. Technological difficulties in data integration, device compatibility, and safe communication exist when integrating blockchain with IoT devices [68]. Robust cryptographic algorithms and secure communication protocols are needed to ensure the integrity and validity of IoT-generated data on the blockchain while preventing manipulation or tampering.

6.6 Smart Contract Security

Supply chain management, temperature monitoring, payment settlements, and other vaccine distribution procedures are automated by smart contracts, self-executing programs installed

on blockchain networks [69]. However, security risks are associated with smart contracts, such as malicious attacks, bugs, and vulnerabilities. To stop potential attacks and guarantee the integrity of vaccination distribution processes, smart contracts must be rigorously tested, subjected to code audits, and have their security and dependability upheld.

To fully utilize blockchain technology in vaccine distribution, interdisciplinary cooperation, creative problem-solving, and ongoing research and development are necessary to address these technological obstacles. To improve transparency, efficiency, and confidence in international healthcare supply chains, stakeholders can develop strong, scalable, and secure blockchain-enabled vaccination distribution systems by overcoming these obstacles.

7 Case Studies

We have also explored different case studies of blockchain applications for vaccinations. Four significant blockchain-enabled case studies for vaccine supply chain management are illustrated in Fig. 20. The detailed studies are as follows:

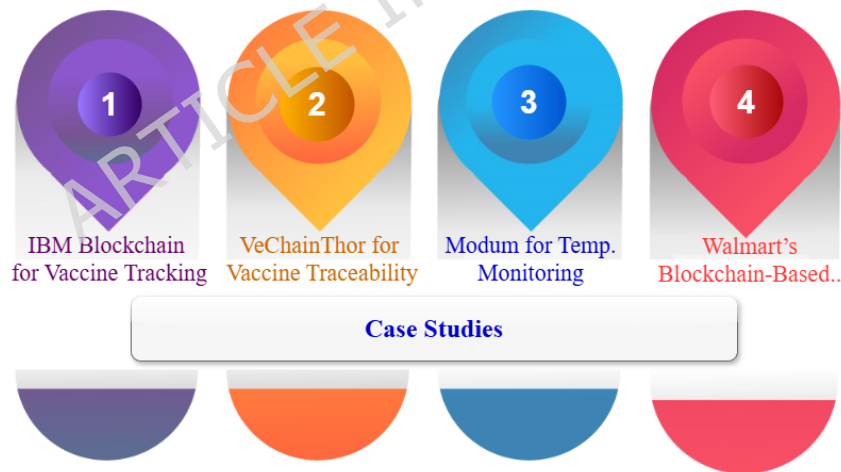


Fig. 20. Blockchain enabled Case studies of the Vaccines Supply Chain

- **IBM Blockchain for Vaccine Tracking:** IBM has developed blockchain-based systems for tracking and distributing vaccines. In 2018, IBM and the US Food and Drug

Administration (FDA) worked together on a pilot project to improve the transparency and security of the supply chain for medications, including vaccines [70]. The project aimed to trace vaccines from producer to patient using blockchain technology, guaranteeing authenticity and keeping fake medications out of the supply chain.

- **China Uses VeChainThor for Vaccine Traceability:** In China, vaccine traceability has been implemented using the blockchain platform VeChainThor. VeChainThor allowed vaccine data to be verified by manufacturers and patients by collaborating with DNV GL, a multinational quality assurance and risk management organization [71]. The production, storage, and delivery of vaccines could all be tracked publicly and securely using blockchain technology, which would increase public confidence in the vaccine supply chain.
- **Modum for Temperature Monitoring:** Vaccines and other temperature-sensitive goods can be tracked during transit with the help of Modum, an IoT and blockchain startup based in Switzerland [73]. Integrating blockchain technology with IoT sensors, Modum guarantees that vaccines stay within the prescribed temperature range during distribution. This keeps vaccinations from spoiling and ensures their effectiveness when they reach their intended location.
- **Walmart's Blockchain-Based Food and Vaccine Safety Pilot Program:** Walmart's blockchain pilot project has ramifications for vaccine delivery, although it primarily focuses on food safety. Walmart hopes to improve supply chain transparency and traceability by utilizing blockchain technology to track food products from farms to store shelves [74]. Similar concepts could be applied to the distribution of vaccines to ensure that vaccines are handled correctly and reach their intended users safely.

8 Future Directions

The development of blockchain technology and its combination with other cutting-edge technologies will determine how vaccines are distributed in the future. Stakeholders can fully utilize blockchain technology to improve transparency, efficiency, and equity in vaccine distribution systems by addressing important research directions like supply chain optimization, cold chain management,

cross-border distribution, data privacy, integration with emerging technologies, and policy considerations. Blockchain technology will be crucial in influencing the future of vaccination delivery and enhancing global health outcomes through interdisciplinary cooperation and innovation. The point-wise future directions are as follows:

- **Combining Blockchain Technology with New Technologies (IoT, AI) to Improve Vaccine Distribution:** Utilizing IoT gadgets to gather data in real-time on vaccine storage conditions and transit routes, such as temperature sensors and GPS trackers. Supply chain interruptions may be avoided [79], demand can be forecast, and vaccination distribution routes can be optimized by applying AI algorithms to analyze data gathered from blockchain networks and IoT devices.
- **Investigating Blockchain Network Interoperability to Promote Cross-Border Vaccine Distribution:** Investigating standards and protocols to enable smooth cross-border data transfer for vaccination distribution across various blockchain networks. Creating cross-chain communication protocols to guarantee data integrity and compatibility while moving information about vaccines between different blockchain platforms.
- **Creating interoperable standards and protocols for blockchain-based vaccine distribution platforms:** Establishing worldwide standards and protocols for blockchain-based vaccination delivery platforms in partnership with governments, industry players, and international organizations. Creating data formats and smart contracts compatible with one another enables smooth data sharing and communication between various vaccine supply chain players.
- **Investigating Scalability Methods for Blockchain Networks to Handle the High Volume of Vaccine Transactions:** Sharding and sidechains are two innovative consensus techniques being explored to help blockchain networks scale and manage the high volume of daily transactions. Investigating layer-two protocols and state channels as off-chain scaling options to ease network congestion and lower blockchain transaction costs.

- **Examining Blockchain-Based Vaccine Distribution Affects Global Health Equity Socio-Economically:** Conduct empirical research to evaluate blockchain-based vaccination distribution's socioeconomic effects on boosting health outcomes, decreasing inequities, and expanding vaccine access in underprivileged areas. The prospective effects of blockchain-enabled vaccine distribution on public health policies, insurance, and healthcare financing should be examined to support fair vaccination distribution worldwide.
- **Frameworks and Policy Consequences for Vaccine Distribution Systems Assisted by Blockchain:** Analyzing the legal environment around blockchain technology and the delivery of vaccines to find adoption potential and roadblocks and having discussions with legislators, regulators, and industry stakeholders to create legislative frameworks for blockchain-enabled vaccination delivery systems that strike a balance between innovation and safeguarding consumer privacy, data security, and protection.

9 Conclusion

In this paper, we have proposed a privacy-preserving vaccine supply chain management. Leveraging blockchain and SSI solution for vaccine fraud reduction, which is caused by its utilization and not keeping a record of vaccine vials (a small glass or plastic vaccine container). The SSI can be traced with privacy preservation from the production unit to the vaccine recipient. Empirical results proved the proposed framework's efficacy in a decentralized, secure, accountable, traceable, transparent, auditable, and trustworthy manner. In addition, the results also justified using a private Ethereum network by performing a cost analysis, using gas, and determining latency to demonstrate the costs associated with using an Ethereum network. Moreover, we also identified and discussed critical challenges current systems face, including scalability issues, interoperability constraints, and privacy concerns. Finally, the findings provided a robust foundation for future research and development in integrating the SSI & blockchain field and achieved potential improvements and strategies to overcome the identified challenges.

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Conflict of interest

The work is not submitted in any other journal. There is no conflict of interest.

Author contribution

The idea and problem formulation along with proposed solution, result analysis, is done by Brijesh Kumar Chaurasia and Nikhil Pateria, verified by all other authors (Brijesh Kumar Chaurasia, Nikita Tiwari, and Nikhil Pateria).

Data availability statement

The dataset generated and/or analyzed during the current study is available upon reasonable request from the corresponding author.

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